Functions: Composite & Inverse

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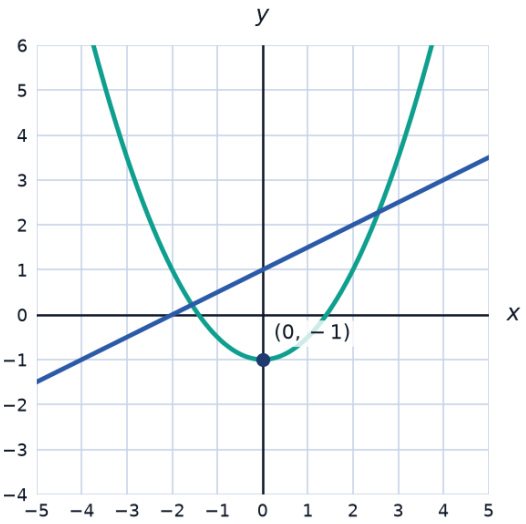
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Domain, Range & Function Concepts

The function $f(x) = \frac{1}{x+6}$ is undefined at one value of x. Find this value.

2 [Foundational] • Target: Composite Functions

The function $f(x) = \frac{1}{x+5}$ is undefined at one value of x. Find this value.



3 [Foundational] • Target: Inverse Functions & Self-Inverses

The function $f(x) = \frac{1}{x+4}$ is undefined at one value of x. Find this value.

4 [Foundational] • Target: Graphical Relationships & Exam Challenges

The function $f(x) = \frac{1}{x+3}$ is undefined at one value of x. Find this value.

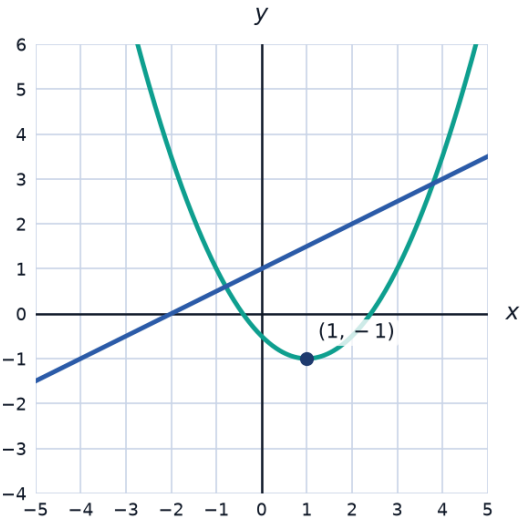
5 [Foundational] • Target: Domain, Range & Function Concepts

The function $f(x) = \frac{1}{x+2}$ is undefined at one value of x. Find this value.

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6 [Foundational] • Target: Composite Functions

The function $f(x) = \frac{1}{x+7}$ is undefined at one value of x. Find this value.



7 [Foundational] • Target: Inverse Functions & Self-Inverses

The function $f(x) = \frac{1}{x+9}$ is undefined at one value of x. Find this value.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Graphical Relationships & Exam Challenges

The function $f(x) = \frac{1}{x+8}$ is undefined at one value of x. Find this value.

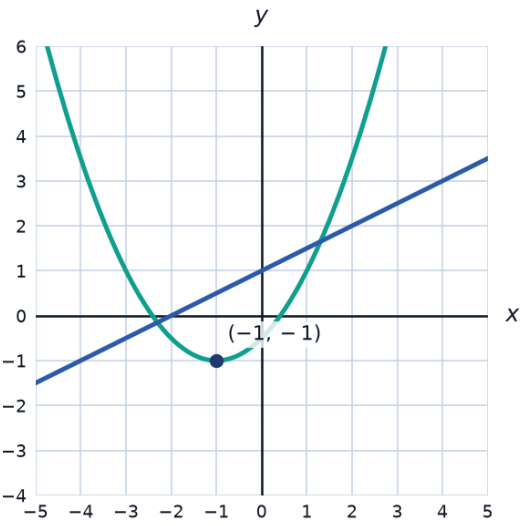
9 [Intermediate] • Target: Domain, Range & Function Concepts

The function $f(x) = \frac{1}{x-4}$ is undefined at one value of x. Find this value.

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10 [Intermediate] • Target: Composite Functions

The function $f(x) = \frac{1}{x-8}$ is undefined at one value of x. Find this value.



11 [Intermediate] • Target: Inverse Functions & Self-Inverses

The function $f(x) = \frac{1}{x-1}$ is undefined at one value of x. Find this value.

12 [Intermediate] • Target: Graphical Relationships & Exam Challenges

The function $f(x) = \frac{1}{x-2}$ is undefined at one value of x. Find this value.

13 [Intermediate] • Target: Domain, Range & Function Concepts

The function $f(x) = \frac{1}{x-3}$ is undefined at one value of x. Find this value.

14 [Intermediate] • Target: Composite Functions

The function $f(x) = \frac{1}{x-7}$ is undefined at one value of x. Find this value.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

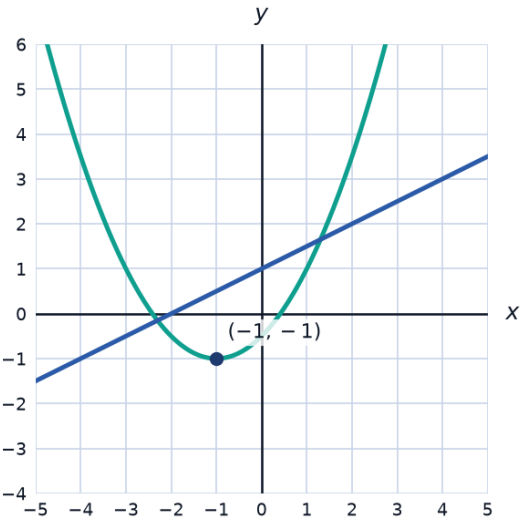
15 [Challenging] • Target: Inverse Functions & Self-Inverses • Cross-Lesson: 4.01 Coordinate Geometry

The function $f(x) = \frac{1}{x-5}$ is undefined at one value of x. Find this value.

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16 [Challenging] • Target: Graphical Relationships & Exam Challenges • Cross-Lesson: 5.01 Trigonometry

The function $f(x) = \frac{1}{x-6}$ is undefined at one value of x. Find this value.



17 [Challenging] • Target: Domain, Range & Function Concepts • Cross-Lesson: 6.01 Differentiation & Integration

Given $f(x) = 5x - 1$ and $g(x) = 2x - 5$, find $gf(x)$.

18 [Challenging] • Target: Composite Functions • Cross-Lesson: 7.01 Statistics & Probability

Given $f(x) = 2x - 5$ and $g(x) = 5x - 2$, find $fg(x)$.

19 [Challenging] • Target: Inverse Functions & Self-Inverses • Cross-Lesson: 9.01 Circular Measure

Given $f(x) = 2x + 2$ and $g(x) = 2x + 6$, find $fg(x)$.

20 [Challenging] • Target: Graphical Relationships & Exam Challenges • Cross-Lesson: 1.01 Indices, Surds & Quadratics

Given $f(x) = 6x$ and $g(x) = 6x + 4$, find $fg(x)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Domain, Range & Function Concepts

Find the natural domain and range of the function $f(x) = \sqrt{2x - 6}$.

a. Domain: $x \geq 3$; Range: $f(x) \geq 0$

b. Domain: $x > 3$; Range: $f(x) > 0$

c. Domain: $x \geq 0$; Range: $f(x) \geq -6$

d. Domain: $x \in \mathbb{R}$; Range: $f(x) \geq 0$

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22 [Foundational] • Target: Domain, Range & Function Concepts

The function $g(x) = x^2 - 4x + 7$ is defined for $x \in \mathbb{R}$. What is the range of g?

a. $g(x) \geq 3$

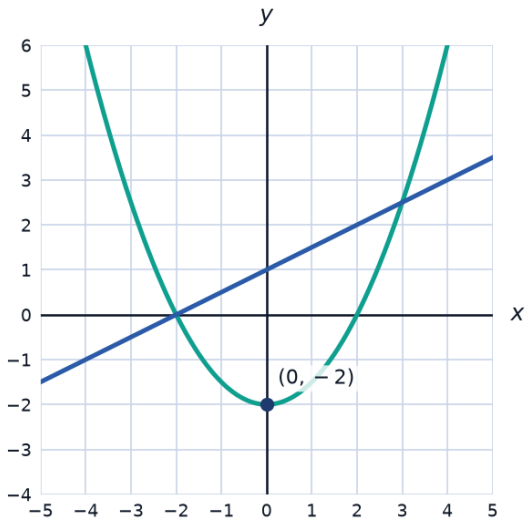
b. $g(x) \geq 7$

c. $g(x) \geq -3$

d. $g(x) \in \mathbb{R}$

23 [Foundational] • Target: Domain, Range & Function Concepts

Explain why the function $h(x) = x^2 - 4$ for $x \in \mathbb{R}$ does not have an inverse.



a. It is a many-to-one function on $\mathbb{R}$

b. Its domain does not include negative numbers

c. Its range is bounded below by -4

d. It is not continuous at $x = 0$

24 [Foundational] • Target: Composite Functions

Given $f(x) = 2x + 3$ and $g(x) = x^2 - 1$, find the expression for $fg(x)$.

a. $2x^2 + 1$

b. $4x^2 + 12x + 8$

c. $2x^2 + 5$

d. $2x^2 - 2$

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25 [Foundational] • Target: Composite Functions

Given $f(x) = \frac{1}{x+2}$ for $x \neq -2$ and $g(x) = 3x - 1$, solve the equation $gf(x) = 1$.

a. $x = 1/2$

b. $x = -1$

c. $x = 2$

d. $x = 0$

26 [Foundational] • Target: Composite Functions

If $f(x) = 3x - 4$, find the expression for $ff(x)$.

a. $9x - 16$

b. $9x - 12$

c. $6x - 8$

d. $9x^2 - 24x + 16$

27 [Foundational] • Target: Inverse Functions & Self-Inverses

Find the inverse function $f^{-1}(x)$ for $f(x) = \frac{2x-5}{x+1}$ $(x \neq -1)$.

a. $f^{-1}(x) = \frac{x+5}{2-x}$

b. $f^{-1}(x) = \frac{x-5}{2-x}$

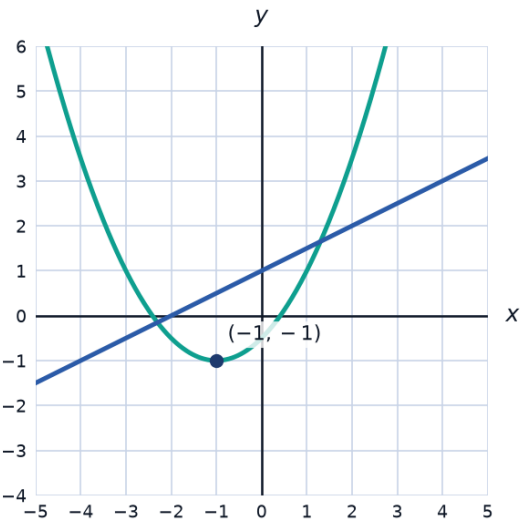
c. $f^{-1}(x) = \frac{2x+1}{x-5}$

d. $f^{-1}(x) = \frac{x+1}{2x-5}$

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28 [Foundational] • Target: Inverse Functions & Self-Inverses

The function $f(x) = x^2 - 6x + 10$ is defined for $x \geq 3$. Find the expression for $f^{-1}(x)$.



a. $f^{-1}(x) = 3 + \sqrt{x - 1}$ for $x \geq 1$

b. $f^{-1}(x) = 3 - \sqrt{x - 1}$ for $x \geq 1$

c. $f^{-1}(x) = \sqrt{x + 6} - 3$

d. $f^{-1}(x) = 3 + \sqrt{x + 1}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Inverse Functions & Self-Inverses

A function $g(x) = \frac{ax + b}{cx + d}$ is a self-inverse function ($g(x) = g^{-1}(x)$). Which condition guarantees this property for all valid x?

a. $a + d = 0$

b. $ad - bc = 1$

c. $a = d$

d. $b = c$

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30 [Intermediate] • Target: Graphical Relationships & Exam Challenges

What is the geometric relationship between the graph of $y = f(x)$ and the graph of $y = f^{-1}(x)$?

a. Reflection in the line $y = x$

b. Reflection in the x-axis

c. Reflection in the y-axis

d. Rotation of 180° about the origin

31 [Intermediate] • Target: Graphical Relationships & Exam Challenges

Solve the equation $f(x) = f^{-1}(x)$ where $f(x) = 2x - 3$.

a. $x = 3$

b. $x = 1$

c. $x = -3$

d. $x = 2$

32 [Intermediate] • Target: Graphical Relationships & Exam Challenges

The function $f(x) = 2x^2 - 8x + c$ has domain $x \geq k$. Find the least value of k for which f has an inverse.

a. $k = 2$

b. $k = 4$

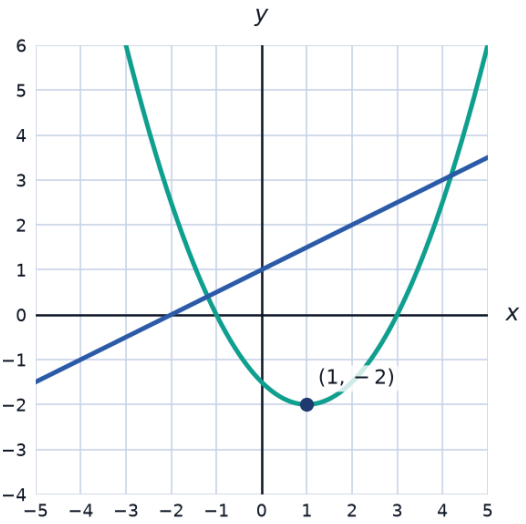
c. $k = 0$

d. $k = -2$

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33 **[Intermediate]** • **Target: Domain, Range & Function Concepts**

The diagram shows the graph of the function $y = f(x)$. The function $f(x)$ is defined for all real numbers. Which of the following represents the graph of $y = f^{-1}(x)$?



a. Graph A

b. Graph B

c. Graph C

d. Graph D

34 **[Intermediate]** • **Target: Composite Functions**

Given $f(x) = 3x - 6$ and $g(x) = 7x + 3$, find $fg(x)$.

a. 10x-3

b. 21x-3

c. 22x+3

d. 21x+3

35 **[Intermediate]** • **Target: Inverse Functions & Self-Inverses**

Given $f(x) = 6x$ and $g(x) = 4x + 9$, find $fg(x)$.

a. 24x-54

b. 10x+9

c. 24x+54

d. 25x+54

36 **[Intermediate]** • **Target: Graphical Relationships & Exam Challenges**

Given $f(x) = x + 3$ and $g(x) = x - 4$, find $gf(-1)$.

a. -4

b. 2

c. -2

d. -1

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37 [Intermediate] • Target: Domain, Range & Function Concepts

Given $f(x) = 5x - 4$ and $g(x) = x + 2$, find $gf(x)$.

a. 6x-2

b. 5x+2

c. 4x-2

d. 5x-2

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Composite Functions • Cross-Lesson: 7.01 Statistics & Probability

Given $f(x) = 2x + 3$ and $g(x) = 4x - 5$, find $fg(x)$.

a. 6x-2

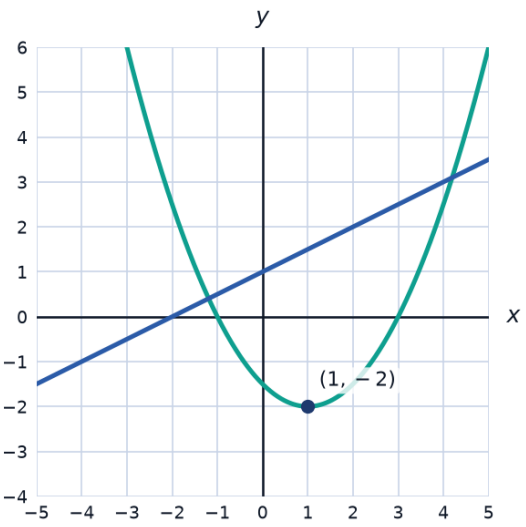
b. 9x-7

c. 8x-7

d. 8x+7

39 [Challenging] • Target: Inverse Functions & Self-Inverses • Cross-Lesson: 9.01 Circular Measure

Given $f(x) = 2x - 5$ and $g(x) = x + 1$, find $gf(x)$.



a. 3x-4

b. 2x-4

c. 2x+4

d. x-4

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40 [Challenging] • Target: Graphical Relationships & Exam Challenges • Cross-Lesson: 1.01 Indices, Surds & Quadratics

Given $f(x) = 2x - 3$ and $g(x) = 2x + 3$, find $fg(x)$.

a. 4x

b. 4x+3

c. 4x-3

d. 5x+3

41 [Challenging] • Target: Domain, Range & Function Concepts • Cross-Lesson: 10.01 Series & Binomial Theorem

Given $f(x) = 5x$ and $g(x) = 2x + 5$, find $fg(3)$.

a. 56

b. -55

c. 11

d. 55

42 [Challenging] • Target: Composite Functions • Cross-Lesson: 11.01 Kinematics

Given $f(x) = x + 5$ and $g(x) = 3x + 1$, find $gf(-1)$.

a. 4

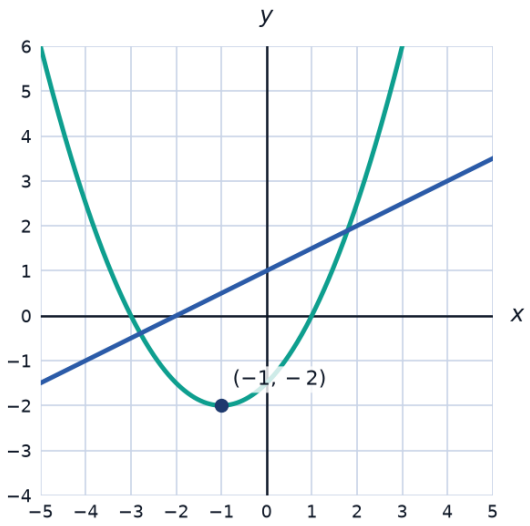
b. 13

c. 14

d. -13

43 [Challenging] • Target: Inverse Functions & Self-Inverses • Cross-Lesson: 12.01 Forces & Equilibrium

Given $f(x) = 2x + 4$ and $g(x) = 3x - 6$, find $gf(x)$.



a. 6x-6

b. 7x+6

c. 6x+6

d. 5x-2

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44 [Challenging] • Target: Graphical Relationships & Exam Challenges • Cross-Lesson: 13.01 Newton's Laws

Given $f(x) = 6x + 9$ and $g(x) = 3x + 5$, find $fg(-1)$.

a. 21

b. -21

c. 2

d. 22

45 [Challenging] • Target: Domain, Range & Function Concepts • Cross-Lesson: 14.01 Energy, Work & Power

Given $f(x) = 3x - 9$ and $g(x) = 4x + 9$, find $fg(-2)$.

a. -5

b. 1

c. -6

d. 6

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